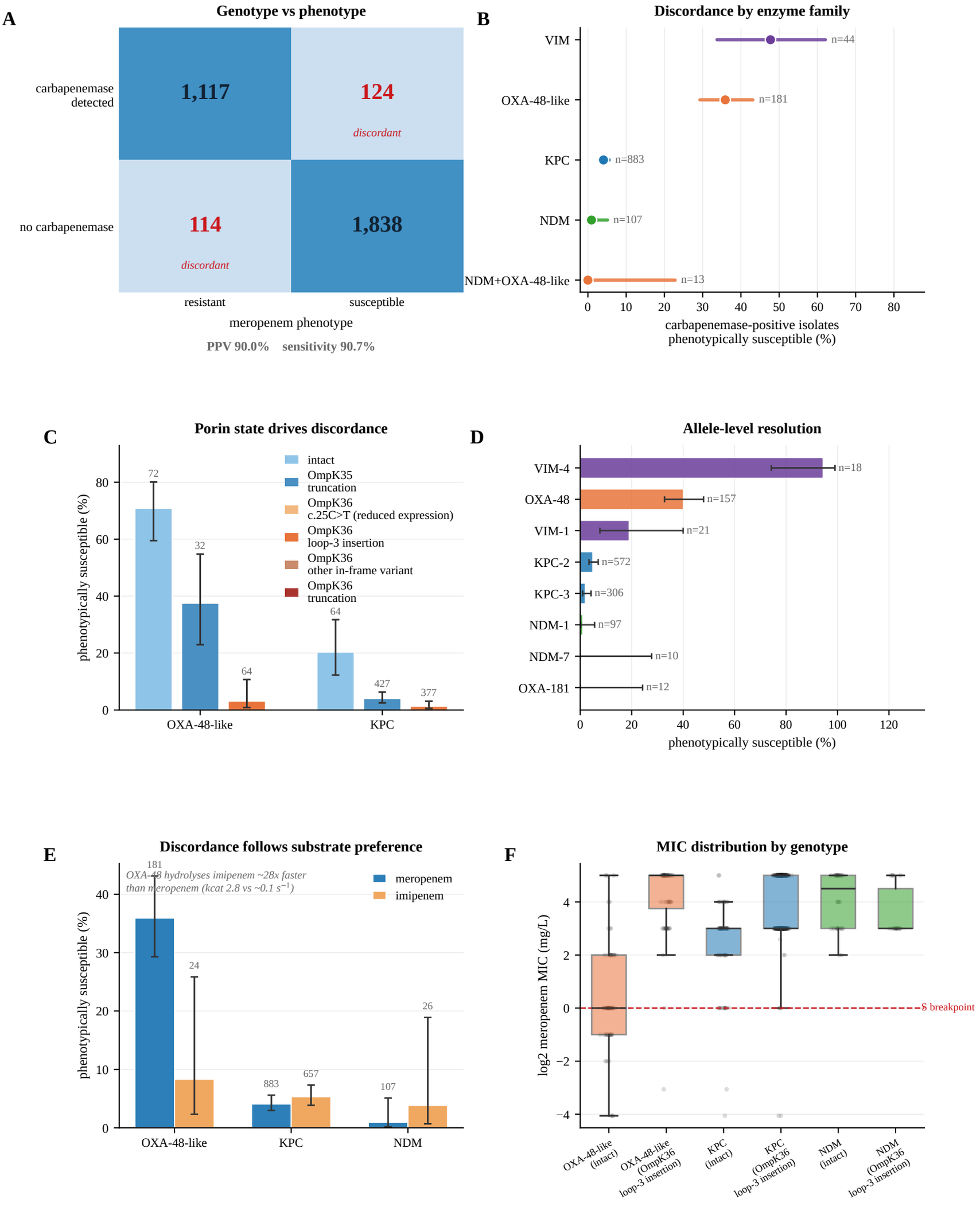
A
Genotype vs phenotype
carbapenemase detected
no carbapenemase
1,117
124
discordant
114
discordant
1,838
resistant
susceptible
meropenem phenotype
PPV 90.0%
sensitivity 90.7%
B
Discordance by enzyme family
VIM
OXA-48-like
KPC
NDM
NDM+OXA-48-like
n=44
n=181
n=883
n=107
n=13
0
10
20
30
40
50
60
70
80
carbapenemase-positive isolates phenotypically susceptible (%)
C
Porin state drives discordance
phenotypically susceptible (%)
intact
OmpK35 truncation
OmpK36 c.25C>T (reduced expression)
OmpK36 loop-3 insertion
OmpK36 other in-frame variant
OmpK36 truncation
72
32
64
64
427
377
OXA-48-like
KPC
D
Allele-level resolution
VIM-4
OXA-48
VIM-1
KPC-2
KPC-3
NDM-1
NDM-7
OXA-181
n=18
n=157
n=21
n=572
n=306
n=97
n=10
n=12
0
20
40
60
80
100
120
phenotypically susceptible (%)
E
Discordance follows substrate preference
OXA-48 hydrolyses imipenem ~28x faster than meropenem (kcat 2.8 vs ~0.1 s⁻¹)
meropenem
imipenem
phenotypically susceptible (%)
181
24
883
657
107
26
OXA-48-like
KPC
NDM
F
MIC distribution by genotype
log2 meropenem MIC (mg/L)
S breakpoint
OXA-48-like (intact)
OXA-48-like (OmpK36 loop-3 insertion)
KPC (intact)
KPC (OmpK36 loop-3 insertion)
NDM (intact)
NDM (OmpK36 loop-3 insertion)